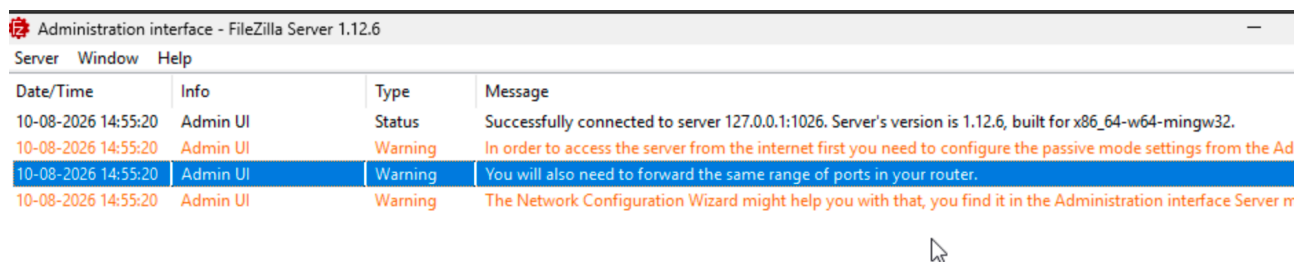


# Module 2: Network + Network Security , Maintenance and Troubleshooting

## Day 4 Other Protocols in Action

### Task 1:



The screenshot shows the 'Administration interface - FileZilla Server 1.12.6' window. It features a menu bar with 'Server', 'Window', and 'Help'. Below the menu is a log table with four columns: 'Date/Time', 'Info', 'Type', and 'Message'. The log contains four entries, all dated '10-08-2026 14:55:20' and from 'Admin UI'. The first entry is a 'Status' message indicating a successful connection to server 127.0.0.1:1026. The next three entries are 'Warning' messages providing instructions on how to access the server from the internet, including configuring passive mode settings and forwarding ports in a router. The third warning message is highlighted with a blue background.

| Date/Time           | Info     | Type    | Message                                                                                                           |
|---------------------|----------|---------|-------------------------------------------------------------------------------------------------------------------|
| 10-08-2026 14:55:20 | Admin UI | Status  | Successfully connected to server 127.0.0.1:1026. Server's version is 1.12.6, built for x86_64-mingw32.            |
| 10-08-2026 14:55:20 | Admin UI | Warning | In order to access the server from the internet first you need to configure the passive mode settings from the Ad |
| 10-08-2026 14:55:20 | Admin UI | Warning | You will also need to forward the same range of ports in your router.                                             |
| 10-08-2026 14:55:20 | Admin UI | Warning | The Network Configuration Wizard might help you with that, you find it in the Administration interface Server r   |

### Task 4:

#### Real-World SNMP Scenarios

1. **ISP network monitoring** — Internet Service Providers use SNMP to poll routers, switches, and modems across their network for metrics like interface bandwidth utilization, packet loss, and device uptime, feeding into dashboards that alert engineers before a link saturates or fails.
2. **Campus/enterprise WiFi management** — University or corporate IT teams use SNMP to monitor wireless access points spread across buildings — tracking how many devices are connected to each AP, signal strength, and whether an AP has gone offline — so they can rebalance load or dispatch a technician without physically checking each device.
3. **Data center / large website infrastructure** — Companies running large-scale web services (e.g., cloud providers or CDN operators) use SNMP to monitor server room equipment like switches, UPS (uninterruptible power supplies), and cooling systems, tracking metrics such as temperature, power draw, and switch port errors to maintain uptime for the websites and apps riding on that infrastructure.

## **Task 5:**

|                             | <b>FTP</b>                                                                                | <b>HTTP</b>                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Port Number</b>          | 21 (control), 20 (active data) or dynamic port (passive data)                             | 80 (HTTP), 443 (HTTPS)                                                                                                      |
| <b>Security</b>             | Plaintext by default (credentials and data unencrypted); secure variant is FTPS/SFTP      | Plaintext in HTTP; encrypted via TLS in HTTPS                                                                               |
| <b>Data Transfer Method</b> | Uses two separate channels — one for commands, one for actual file data                   | Single connection carries both the request and the response/data                                                            |
| <b>Use Cases</b>            | Bulk file uploads/downloads, website deployment, transferring large files between servers | Loading web pages, API calls, streaming, general web browsing                                                               |
| <b>Authentication</b>       | Requires a username and password (or allows "anonymous" login) for every session          | Often stateless/no login required for public pages; auth handled separately via cookies, tokens, or login forms when needed |

